

KIT's College of Engineering Kolhapur (Empowered Autonomous)

KIT's Innovation and Research Foundation

KIT's IRF, Kolhapur

Section (8) Company

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And

DBATU Forum of Innovation, Incubation and Enterprise (DFIIE)

Hosted by Dr. Babasaheb Ambedkar Technological University (DBATU)

Pitch Deck Presentation for IDEATHON

VIRO - AI

A New Era of Proactive Health

by

Team Rakshanu



Hosted By



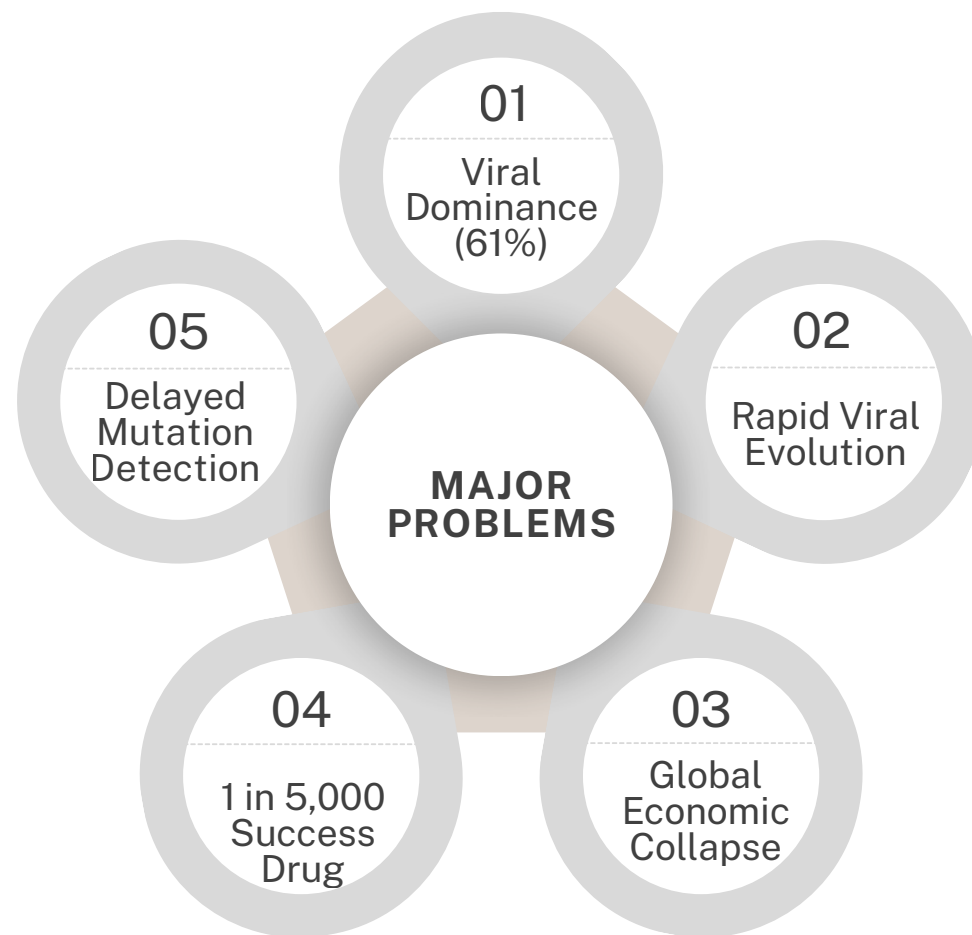
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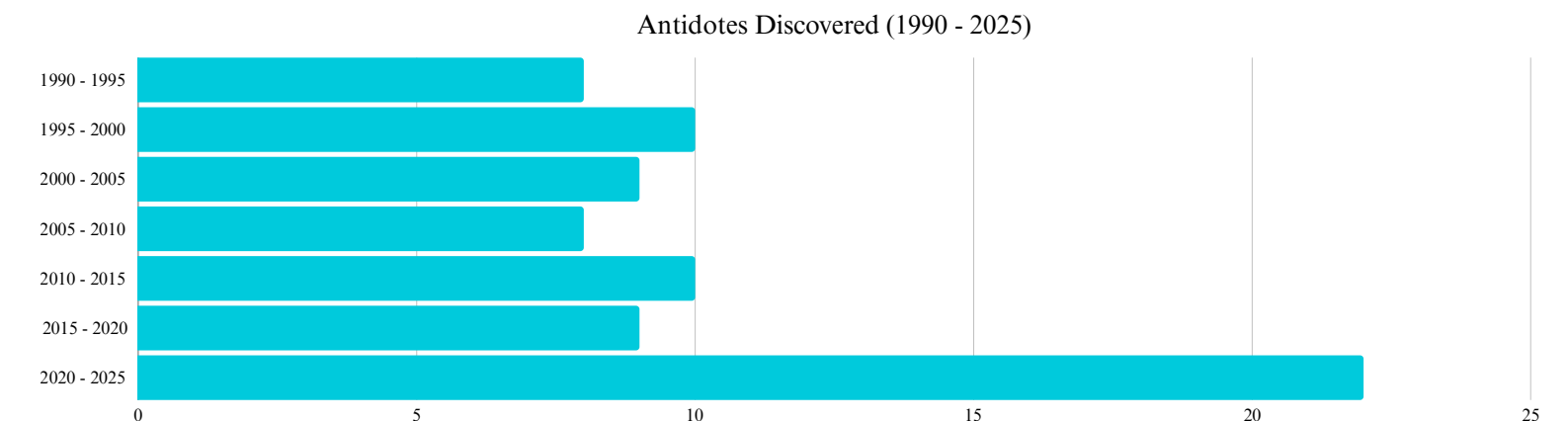
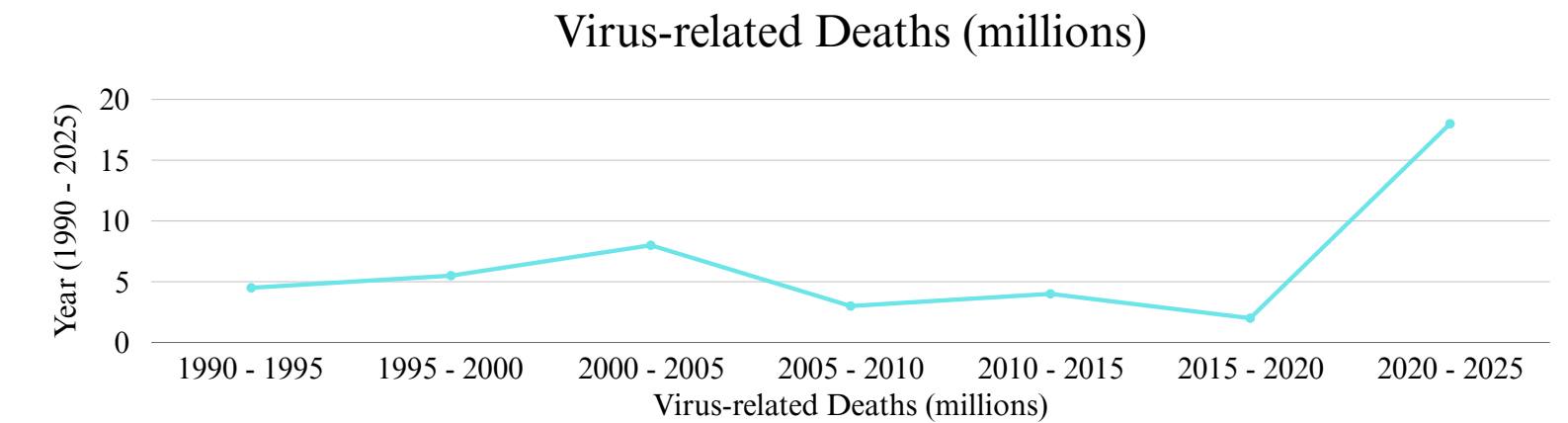
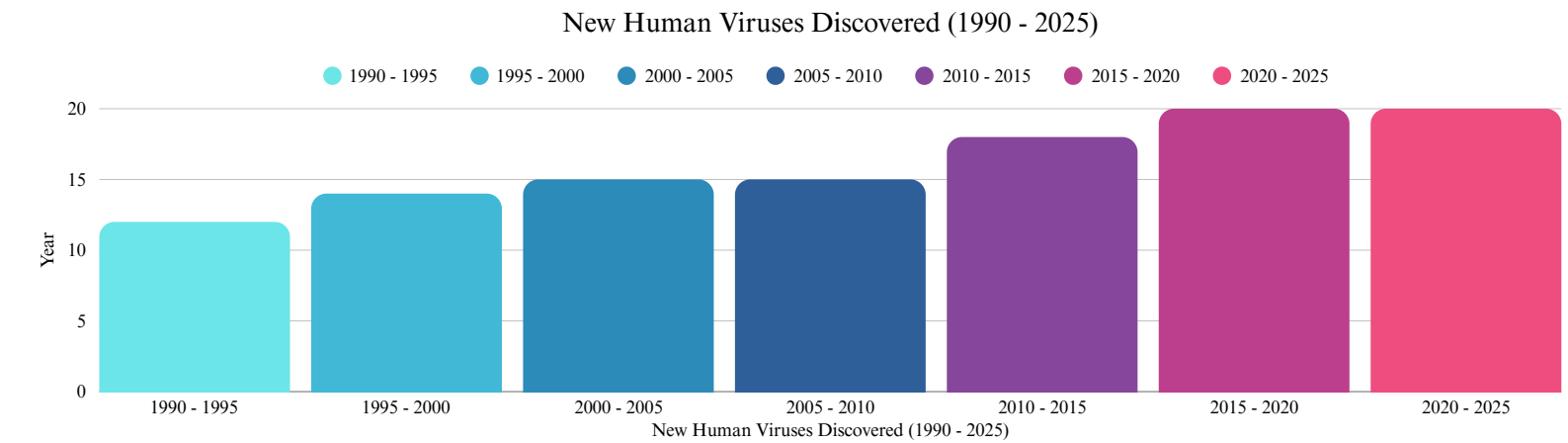


PROBLEM STATEMENT

Rising healthcare needs and growing populations drive demand for innovative biotechnologies in viral prediction and drug discovery.



The Black Death wiped out nearly one-third of Europe's population — 25 to 50 million lives lost in just 7 years."

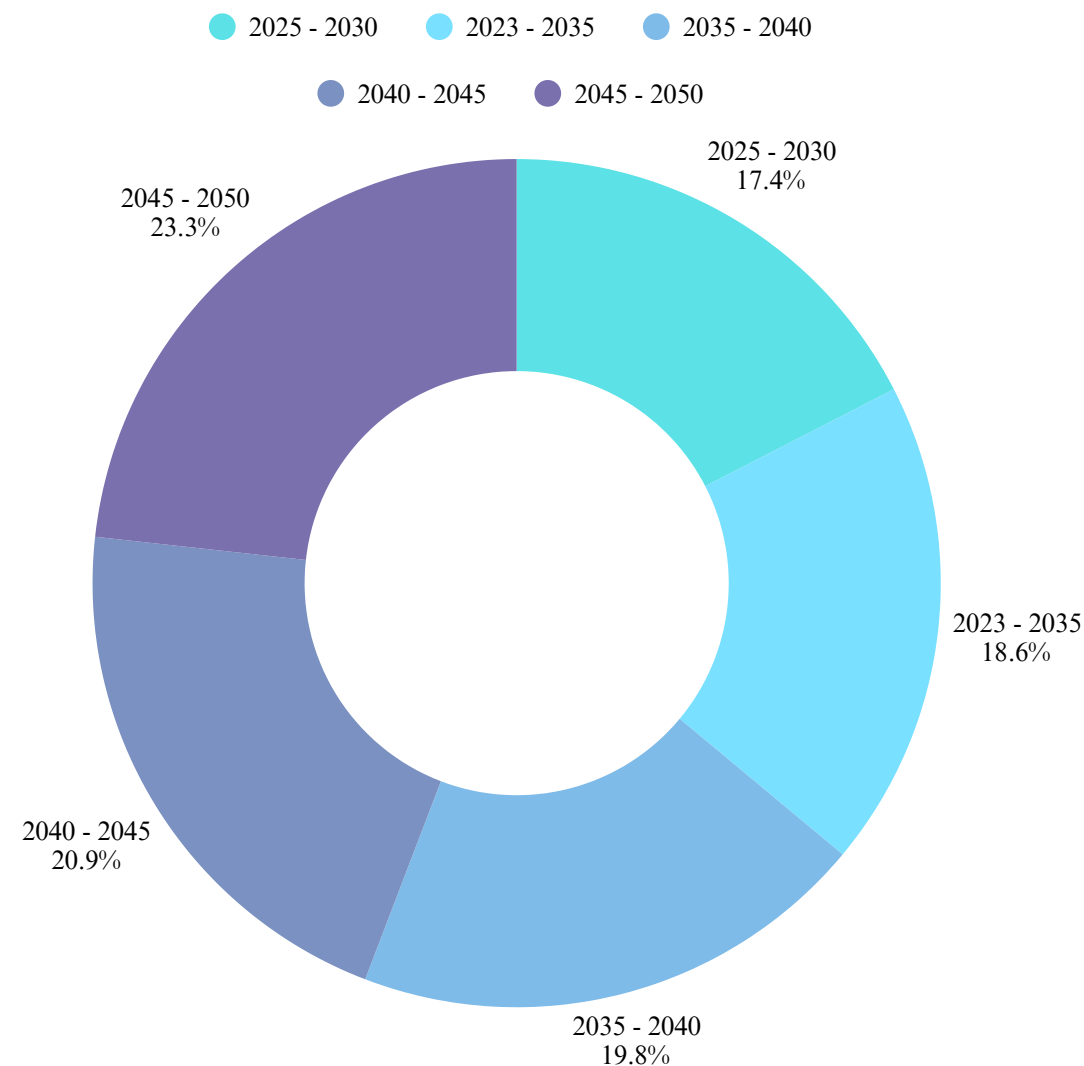




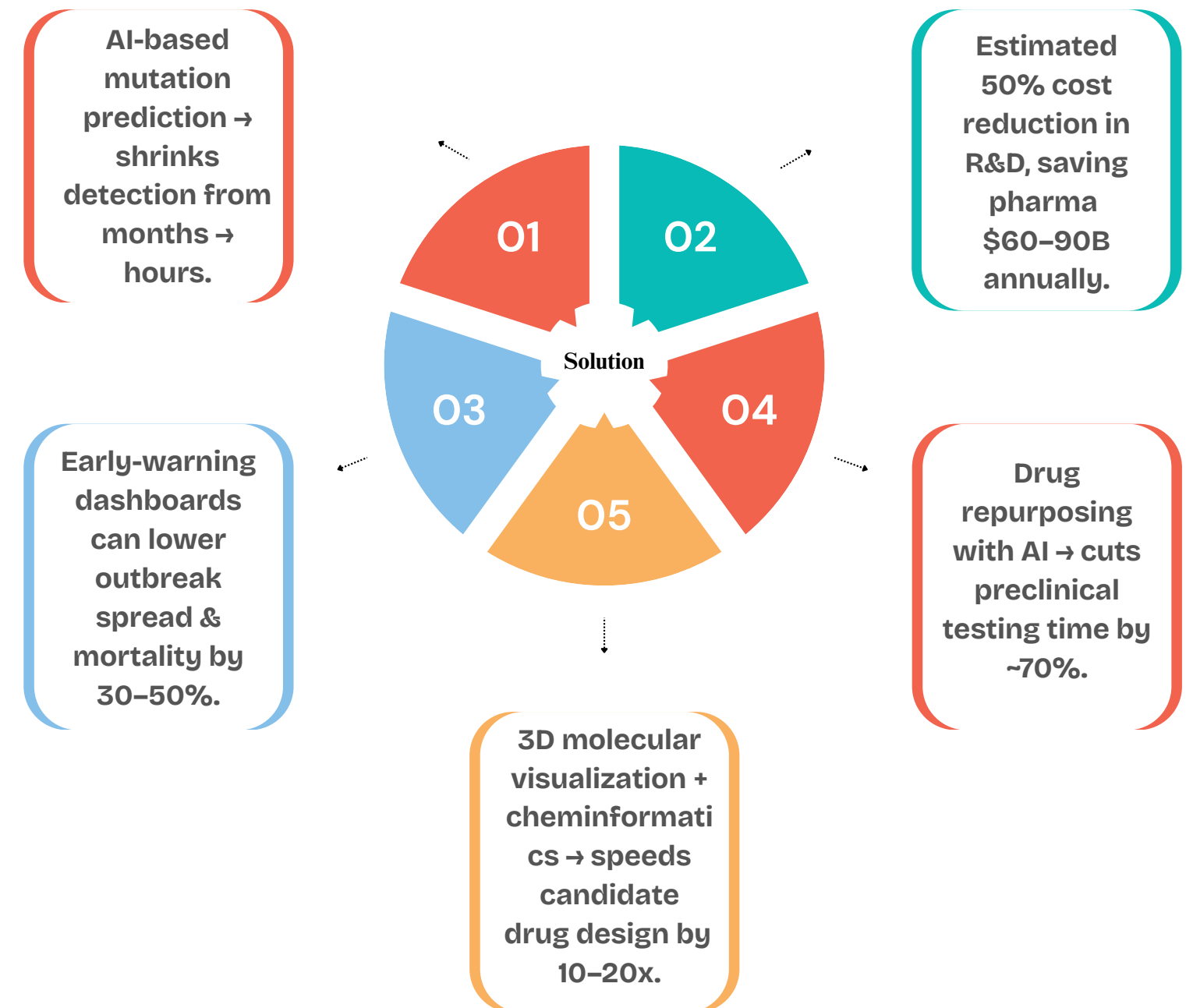
SOLUTION

Viro-AI: An AI-driven bioinformatics platform that forecasts viral mutations, simulates protein structures, and identifies potent drug candidates to accelerate outbreak response and therapeutic discovery.

Our System Accuracy (if implemented)



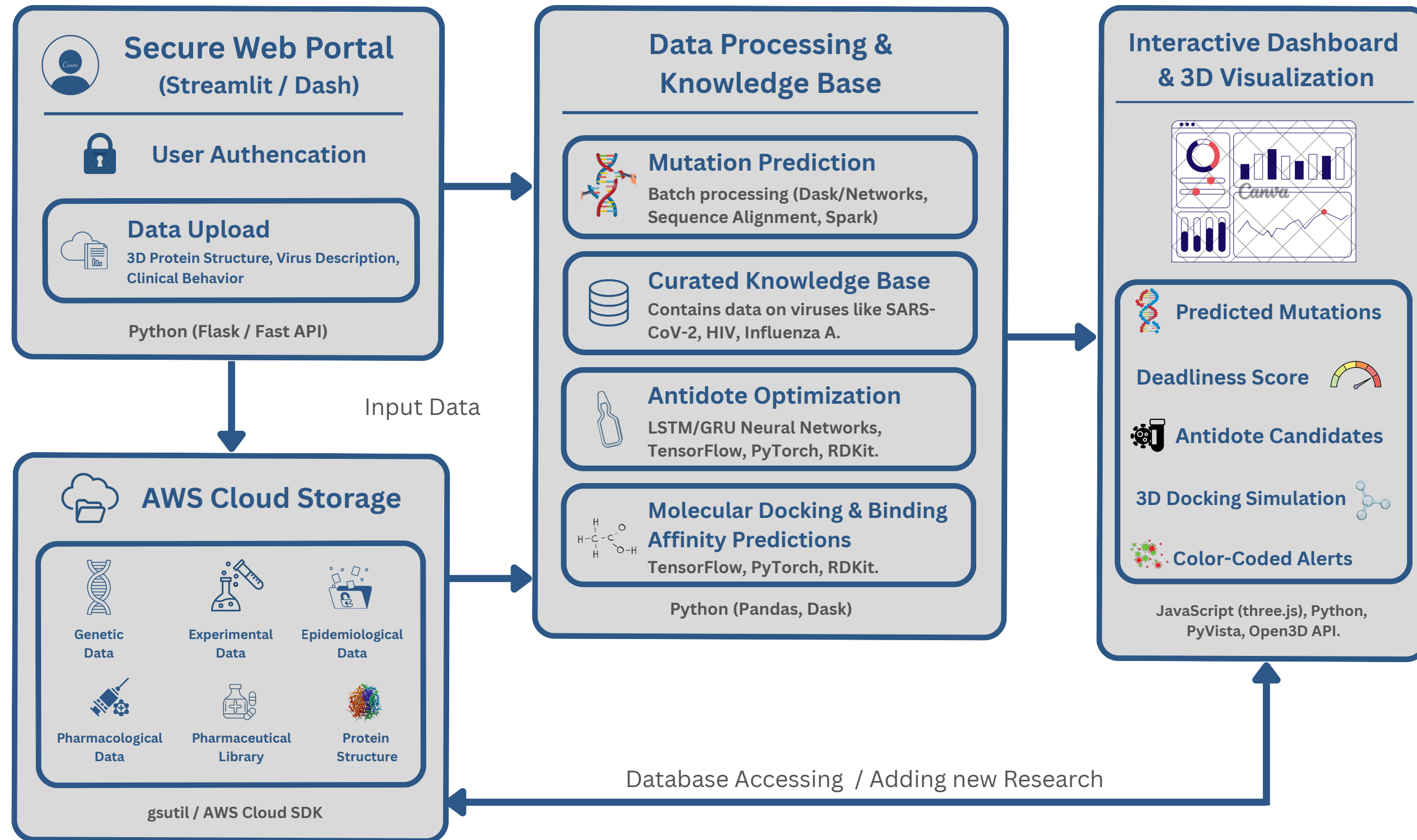
Viral Insight & Rapid Optimization Analytics Intelligence



TECHNICAL APPROACH



SYSTEM ARCHITECTURE





BUSINESS IDEOLOGY

Phase 1

Research & Validation (Years 1-2)

- **Focus :** Establish Scientific Credibility
- **Product :** Research-as-a-Service (RaaS).
- **Target Audience :** University Labs & Biotech Startups.
- **Revenue Model :** Research Grants & Project-Based Contracts.
- **Key Goal :** Achieve >70% prediction accuracy in a published case study and secure foundational research partners.

Phase 2

Commercialization (Years 2-4)

- **Focus :** Build a Scalable Business
- **Product :** Tiered SaaS Platform (Basic, Pro, Enterprise).
- **Target Audience :** Mid-Sized Pharma, CROs, and Health Agencies.
- **Revenue Model :** Recurring Subscriptions & Usage-Based Fees.
- **Key Goal :** Onboard first paying enterprise clients and achieve key regulatory compliance (e.g., HIPAA).

Phase 3

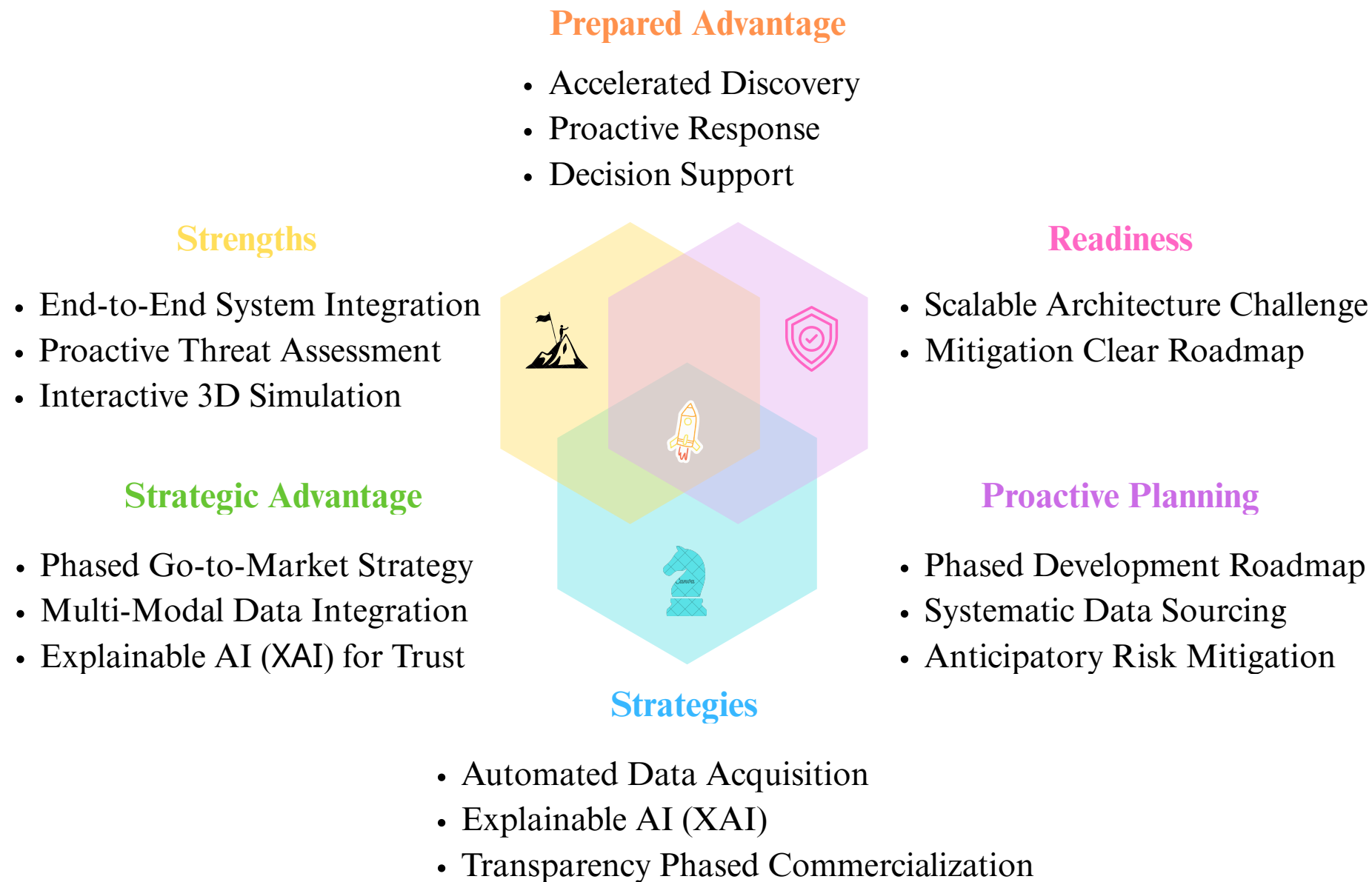
Ecosystem Expansion (Year 5+)

- **Focus :** Become an Industry Standard
- **Product :** Predictive drug discovery and viral mutation analysis Software
- **Target Audience :** Large Pharma, Governments, and Global Health Organizations (WHO).
- **Revenue Model :** Government Contracts & API Marketplace Revenue Sharing.
- **Key Goal :** Integrate into national pandemic preparedness plans.

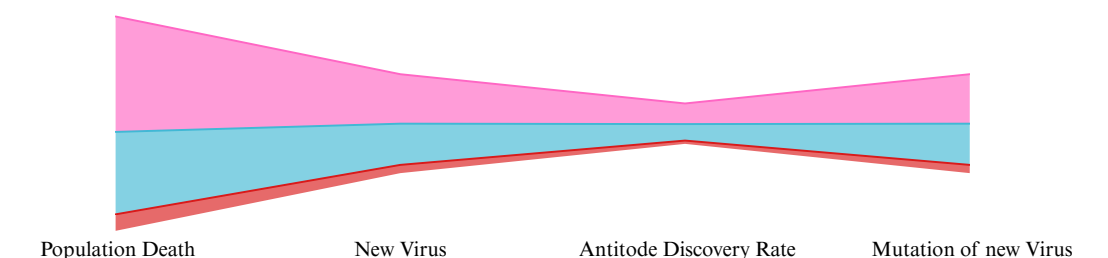
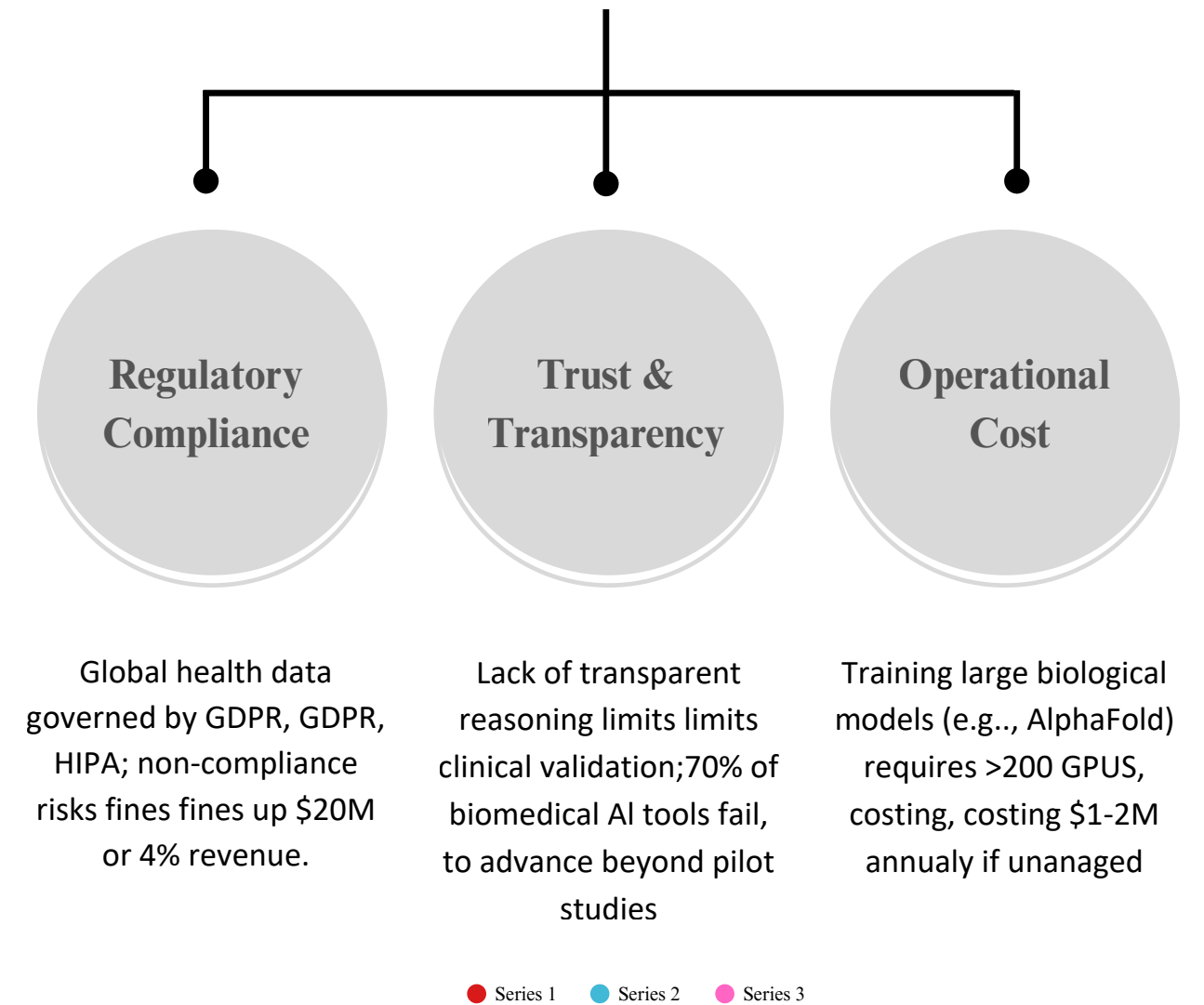


FEASIBILITY & VIABILITY

The Synergy of Strengths, Readiness, and Strategies



Potential Challenges





IMPACT & BENEFITS

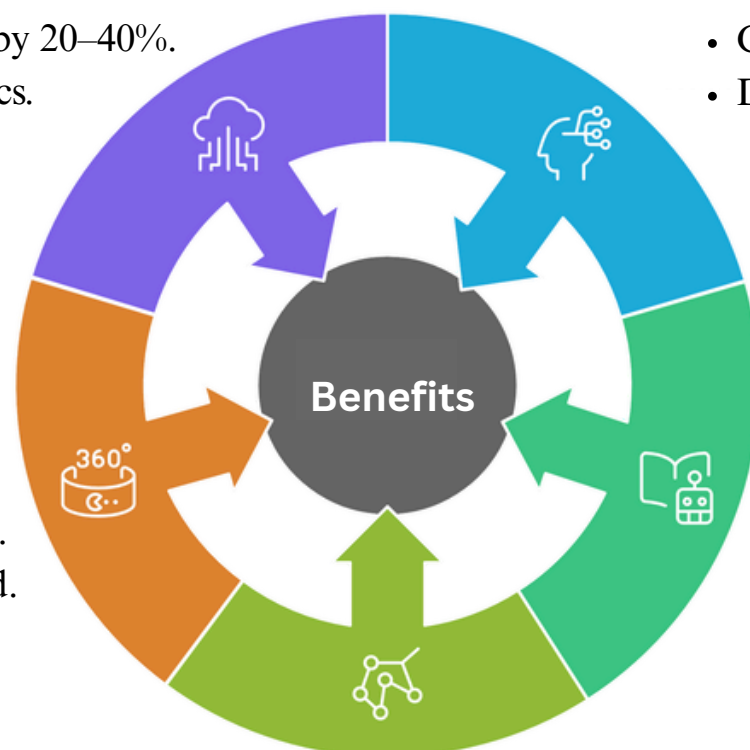
Benefits of VIRO - AI System

Social

- Early detection cuts fatality rates by 20–40%.
- Saves millions of lives in pandemics.

Healthcare

- 3D molecular visuals for clinicians.
- Cuts 80% bioinformatics workload.



Environmental

- Less animal testing with simulations.
- Tracks zoonotic diseases.
- Lower R&D carbon footprint.

Economic

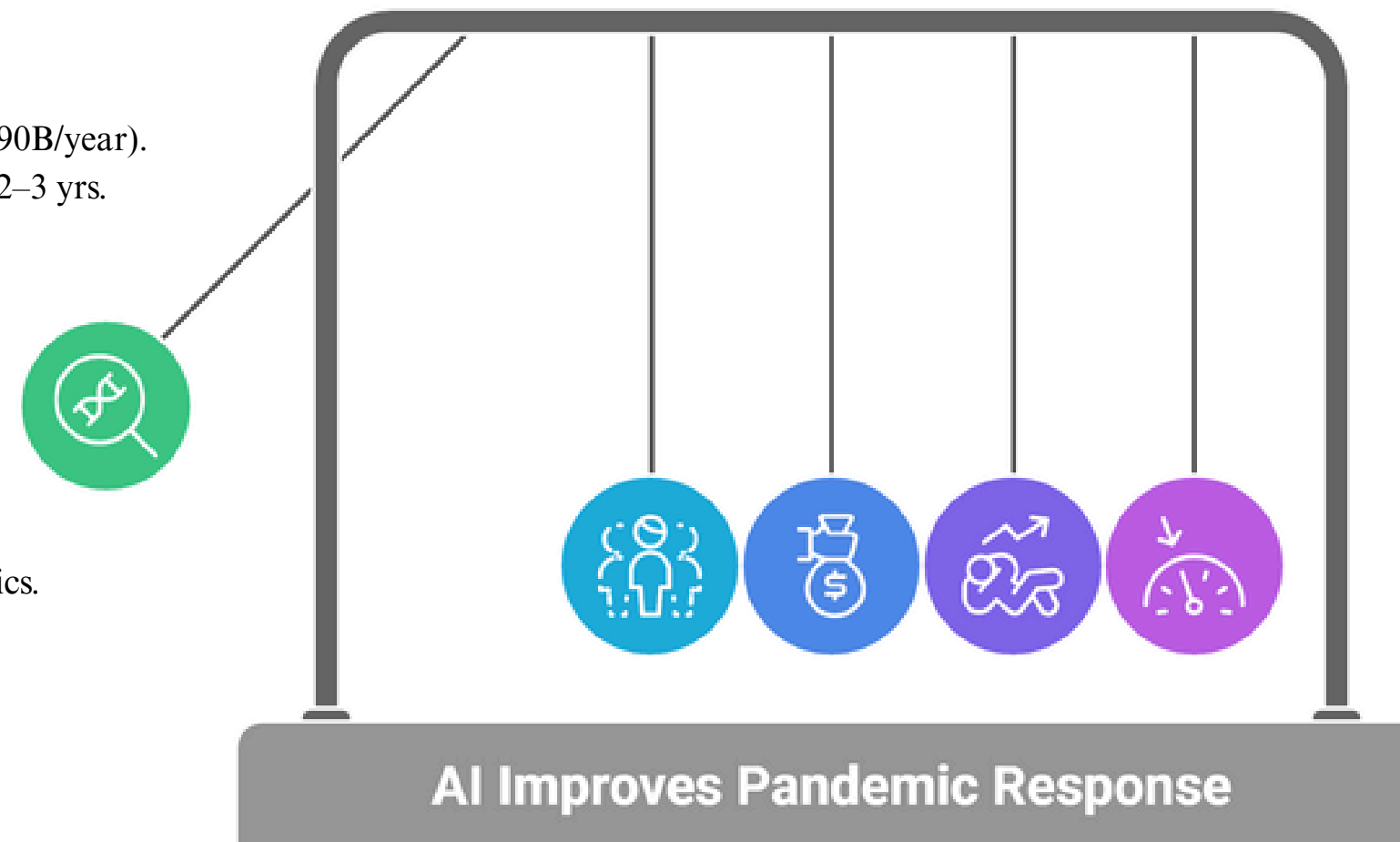
- Cuts R&D costs by 30–50% (saves \$60–90B/year).
- Drug discovery time reduced: 12 yrs → 2–3 yrs.

Technological

- Unified AI + ML + Cheminformatics.
- Cloud-native,
- 99.9% uptime global monitoring.



Impact of Pandemic Preparedness



Reduced Transmission
Up to 60% reduction

Economic Savings
\$4–5T cost reduction

Reduced Fatality
20–40% fatality reduction

Faster Research
Discovery time cut drastically



RESEARCH & REFERENCES

Sr. No	Author	Paper Title	Publication	Year	Link
1)	Jumper, J. et al.	AlphaFold: Highly accurate protein structure prediction	Nature	2021	https://www.nature.com/articles/s41586-021-03819-2
2)	Kokudeva, M. et al.	AI as a tool in drug discovery and development	Biotechnology & Biotechnological Equipment	2021	https://www.nature.com/articles/s41573-019-0016-z
3)	Deng, J. et al.	Artificial intelligence in drug discovery: applications & techniques	Briefings in Bioinformatics	2020	https://www.nature.com/articles/s41573-019-0016-z
4)	Vamathevan, J. et al.	Machine learning in drug discovery and development	Nature Reviews Drug Discovery	2019	https://www.nature.com/articles/s41573-019-0016-z

https://drive.google.com/drive/folders/1BQ701dxHz6_xExREs6aze_4dtTX07OtU?usp=sharing

Project Details Link

<https://viroai.netlify.app>

Prototype Link



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TEAM MEMBERS & ROLE

Team Leader Name : Sairaj Popat Jadhav
Branch : Bachelor of Engineering

Role : Backend Developer, Database Manager, Research
Stream : Artificial Intelligence & Data Science **Year :** TE

Team Member 1 Name : Vedant Lanjekar
Branch : Bachelor of Engineering

Work : Backend Developer, Database Manager, Research
Stream : Artificial Intelligence & Data Science **Year :** TE

Team Member 2 Name : Pranav Nitin Shinde
Branch : Bachelor of Engineering

Work : Frontend Developer, Research
Stream : Computer Engineering **Year :** TE

Team Member 3 Name : Vaishnavi Prakash Patil
Branch : Bachelor of Engineering

Work : Frontend Developer, Research, Documentation
Stream : Computer Engineering **Year :** TE



ACHIEVEMENTS ON THIS PROJECT

Ideathon 2k25

organized by

**Dr. D. Y. Patil College of Engineering & Innovation
Varale, Talegaon, Pune - 410507**

1st Prize (Certificates, Trophy, Cash Prize : Rs: 2000)



Smart India Hakathon (Round 1 : Internal Round)

organized by

**Dr. D. Y. Patil College of Engineering & Innovation
Varale, Talegaon, Pune - 410507**

1st Prize (Certificates, Trophy)





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ANY QUESTIONS ??



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THANK YOU